FIFA

2025 EDITION

FIFA QUALITY PROGRAMME FOR ELECTRONIC PERFORMANCE & TRACKING SYSTEMS TEST MANUAL

VERSION: MAY

Contents

1. Introduction 4

2. Test Protocol 5

2.1. Testing Venue 5

2.2. Pitch dimensions 5

2.3. Test area 5

2.3.1. VICON 3-D Motion Capture System Set-up 6

2.4. Player & Ball Preparation 7

2.4.1. Player Centre of Mass (COM) 7

2.4.2. Player Lower Limb Tracking 8

2.4.3. Player Inertial Measurement Units (IMUs) 9

2.4.4. Ball Preparation 9

2.5. Start & End Times 9

2.6. Player Tracking Protocols 10

2.6.1. Circuits 10

2.6.2. Maximal Sprints 10

2.6.3. 2v2 Small-Sided Games 11

2.6.4. 5v5 Small-Sided Game 11

2.7. 2D & 3D Ball Tracking 11

2.7.1. 10 m, 20 m, and >20 m Passes 11

2.7.2. Ball Dribble – Continuous Tracking 11

2.7.3. Throw-ins 11

2.7.4. Small-Sided Games - Continuous Ball Tracking 12

2.7.5. Full Pitch Stationary Position 12

2.8. Live Tracking Data 13

2.8.1. Server Upload Assessment 13

3. Data Analysis 14

3.1. Provider Data Submission 14

3.1.1. Position & Speed 14

3.1.2. Acceleration 14

3.1.3. Load 15

3.1.4. Ball & Kick Metrics 15

3.2. Measurement Units 15

3.3. 3-D Motion Capture Data Preparation 17

3.4. Data Synchronisation 17

3.5. Position Alignment 17

3.6. Statistical Analysis 18
3.6.1. Root Mean Square Difference 18
3.6.2. Mean Absolute Difference 18
3.6.3. Sensitivity 18

4. Position & Speed - Criteria for Product Accreditation 19
4.1. Position 19
4.2. Speed 19
4.3. Rating System 19

5. New Metrics 20
5.1. Acceleration Evaluation 20
5.1.1. Time-series Acceleration 21
5.1.2. Acceleration Efforts 21
5.2. Load Evaluation 22
5.2.1. Load Index 22
5.3. Ball & Kick Metric Evaluation 23
5.3.1. Ball Contacts & Kick Detection 24
5.3.2. Foot-to-Ball Contact Time 24
5.3.3. Foot Speed – Peak Speed 25
5.3.4. Foot Speed – Mean Foot Speed 25
5.3.5. 2-D Ball Position & Speed 26

6. References 27

Appendix A: Example Product Certification Report (EPTS Test Manual: 2022 Edition) 28

Appendix B: Example Figures for New Metrics 39

Appendix C: Guidelines: EPTS Live Tracking Data Assessment via gRPC 41

1. Introduction

Following extensive research and consultation, FIFA has developed a Quality Programme for Electronic Performance & Tracking Systems (EPTS), covering both player and ball tracking. This programme conducts standardised performance assessments relating to speed and position derived outputs from EPTS used in the football industry and certifies these systems based on their determined quality outcomes. The programme has recently expanded to include the quality evaluation of new football performance metrics: acceleration, load, and technical metrics. These are not eligible for certification in 2025 but, by mutual agreement between the provider and the test institute, they can be included in the reports.

The testing methodology relies on three core systems to evaluate EPTS performance metrics:

1. **3-D Motion Capture (VICON):** An established criterion system to assess football-specific movements (i.e., speed, position, acceleration, and technical metrics) within a designated capture space on the pitch.

2. **Inertial Measurement Units (IMUs):** VICON Blue Trident IMUs as a criterion system for load metric validation.

3. **Pitch Survey (Total Station):** Quantity surveying equipment is used to measure key areas of the pitch to verify system accuracy under match conditions.

This test manual applies to the majority of EPTS types, including Global Positioning Systems (GPS), Global Navigation Satellite System (GNSS), Local Positioning Systems (LPS), Optical Tracking Systems (OTS), Light Detection and Ranging (LiDAR), and Real Time Kinematic (RTK) systems. It outlines the standardised procedures for data collection, processing, and statistical analysis, designed to determine the quality of performance tracking systems used in football.

2. Test Protocol

2.1. Testing Venue

Testing will be undertaken in a football stadium with pitch dimensions according to the official FIFA regulations. The test area may be set up in any pitch location, however, where possible it will utilise the centre of the pitch (see Figure 1) to allow each provider the opportunity to obtain the best possible results. The venue must allow sufficient height for infrastructure for optical player and ball tracking system cameras to be installed.

2.2. Pitch dimensions

Pitch dimensions and coordinates of known locations on the pitch will be surveyed via Total Station for use in subsequent full-pitch analyses. These will be given to providers prior to testing. Providers may also seek to determine dimensions using their own methodology.

2.3. Test area

A 40 x 30 m area for player and ball tasks (Sections 2.6 & 2.7) will be used as the 3-D motion capture space. An example of four potential set-up positions for the 3-D motion capture test area are shown in Figure 1.

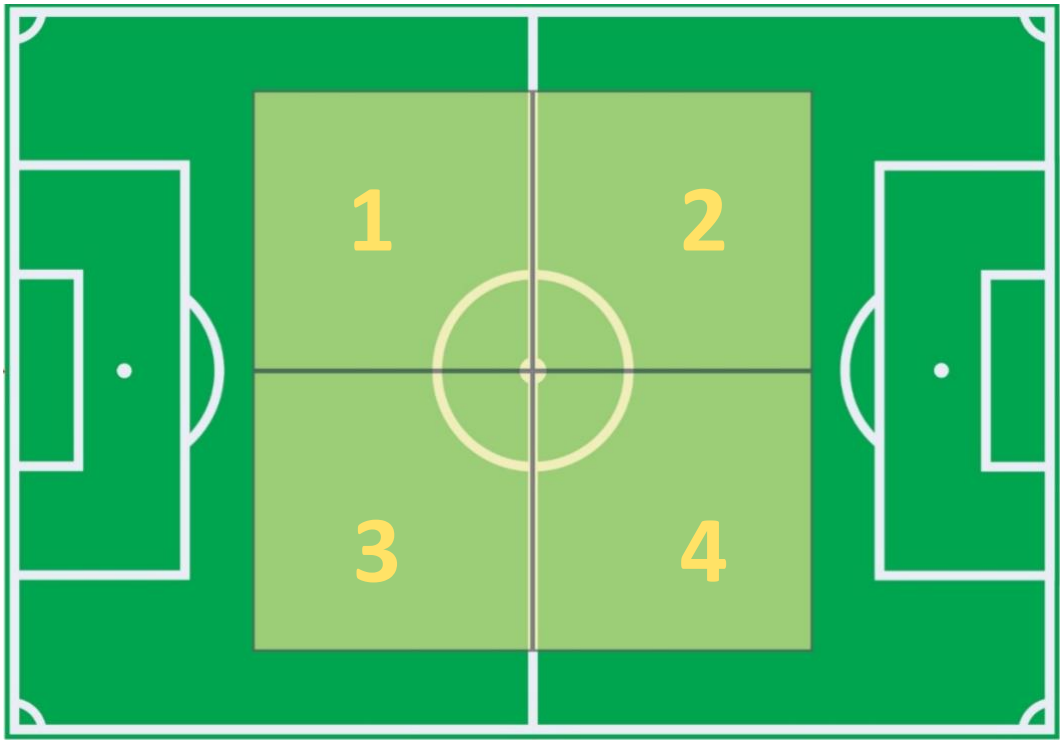


Figure 1. *Typical position of test area on playing pitch. The yellow shaded areas represent four possible 40 x 30 m quadrants, with the motion capture system on its perimeter.*

2.3.1. VICON 3-D Motion Capture System Set-up

Forty VICON cameras (34 x Valkyrie, 6 x Vantage V16) (Oxford Metrics Group Plc [OMG], Oxford, UK) will be evenly spaced around a perimeter of the test area, allowing for a minimum 35 m x 25 m VICON capture space (Figure 2). All VICON trials will be recorded at a sampling rate of 100 Hz. The cameras are powered over ethernet (POE) via switches connected through Cat 6E ethernet cables. All switches are connected to a central hub switch and computer. Valkyrie and Vantage V16 cameras with high megapixel sensors are used as they are suitable for outdoor capture due to their water resistance functionality and fitted optical filter which attenuates wavelengths of other light to only capture a specific infrared wavelength. Following camera set-up, the system will be calibrated, with the test institute verifying calibration error values are within acceptable limits (<1 mm). Re-calibrations will be performed where necessary during the test event to account for environmental changes (e.g. light and temperature) and camera shifts (e.g. accidental tripod or camera bump) that may affect camera accuracy.



Figure 2. *Example of 3-D motion capture system test area set-up*

2.4. Player & Ball Preparation

2.4.1. Player Centre of Mass (COM)

Six reflective markers with a 40 mm diameter will be placed on each participants left and right anterior and posterior superior iliac spine (hips) and left and right shoulders (see Figure 3). Players will wear custom Velcro shorts under their uniform shorts to allow attachment of the pelvic markers. Markers will be adhered to the four pelvic landmarks via Velcro bases, with Velcro straps used to ensure firm fitting of shorts to player hips. Markers will be secured to the shoulders with a combination of double-sided tape between the marker base and the skin with strapping tape placed over the top of the base. The same testers will apply the markers each day to provide consistent marker locations for each participant.



***Figure 3**. Position of the shoulder and pelvic retro-reflective markers*

2.4.2. Player Lower Limb Tracking

Lower limb tracking is used for new metrics for providers capturing ball and kick metrics. Where relevant, players lower limb tracking will be captured by the VICON 3-D motion capture system using 26 x 25 mm diameter retro-reflective markers. These markers will be placed on participants left and right lower limbs prior to testing (see Figure 4). The addition of the lower limb markers in Figure 4 allows joint (i.e. hip, knee, and ankle), and limb segment centre (i.e. thigh, shank, and foot) position and speed to be established. Participants will wear custom Velcro pants, with markers attached and secured via Velcro bases. Thigh and shank segments will be captured using semi-rigid bodies with 3-4 reflective markers attached. Foot markers will be secured with a combination of double-sided tape between the marker base and the football boot, with strapping tape placed on top of the base of the marker.

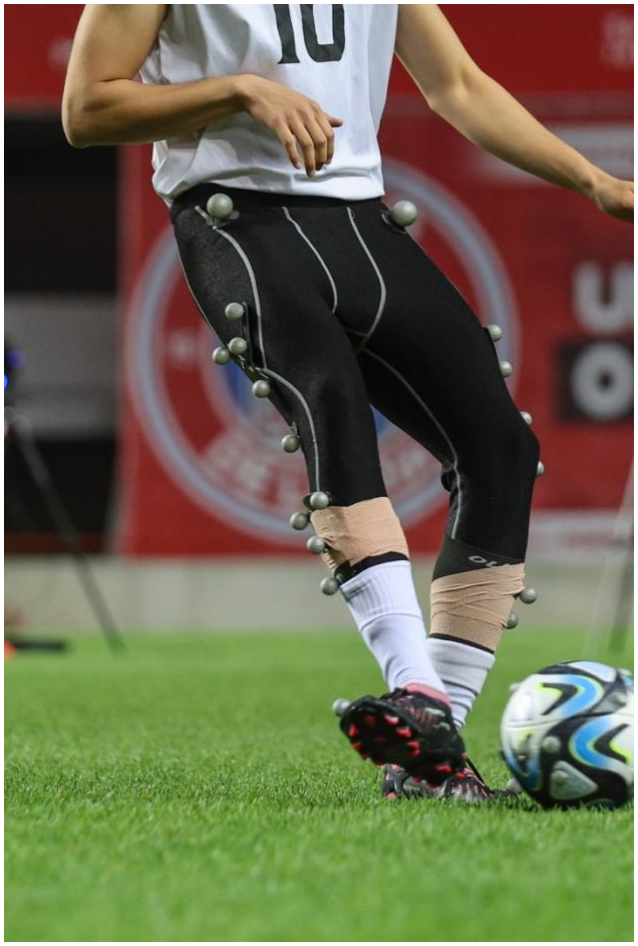


***Figure 4**. Example of lower limb retro-reflective marker placement for VICON tracking.*

2.4.3. Player Inertial Measurement Units (IMUs)

For providers measuring load index, VICON Blue Trident IMUs will be secured to a player's pelvis between the left and right posterior superior iliac spine markers. The IMUs will be attached to custom Velcro shorts using Velcro strips on the back of the device, with additional Velcro straps securing them in place for the test session. Provider's requesting specific placement of the IMUs in line with their system can coordinate with the test institute to facilitate IMU placement. Alternative attachment methods include Velcro to different landmarks, double-sided tape, sports tape, or an IMU limb-mounting band.

2.4.4. Ball Preparation

Ball tracking will be captured in VICON using the mid-point of 16 retro-reflective half-dome markers adhered to the ball (Figure 5). This 16-marker ball model will be used for all ball and kick tracking trials. All VICON ball models used for 2-D and 3-D testing will be created in VICON Tracker software to cross-check ball geometric centre prior to testing.



***Figure 5**. Example of retro-reflective ball integrated into the ball for VICON tracking.*

2.5. Start & End Times

A final check with providers will be carried out before starting the session to ensure that participating providers are ready to begin. Upon provider confirmation, a whistle will be blown to indicate the beginning of the session. Providers must submit data from this point for all players (and ball, if applicable) until the second whistle is blown. Individual VICON trial start and finish times will be recorded to the nearest tenth of a second to assist with temporal synchronisation with provider trials.

2.6. Player Tracking Protocols

For speed, position, acceleration, and load metrics, 10 players will complete the following tasks:

2.6.1. Circuits

Two circuits designed as demonstrated in (Figure 6) will be performed in the VICON capture area, and will include:

i. Self-paced walking

ii. Self-paced jogging

iii. Maximal accelerations

iv. Changes of direction

Each participant shall perform three repetitions of the circuit, with five participants present in each circuit trial.

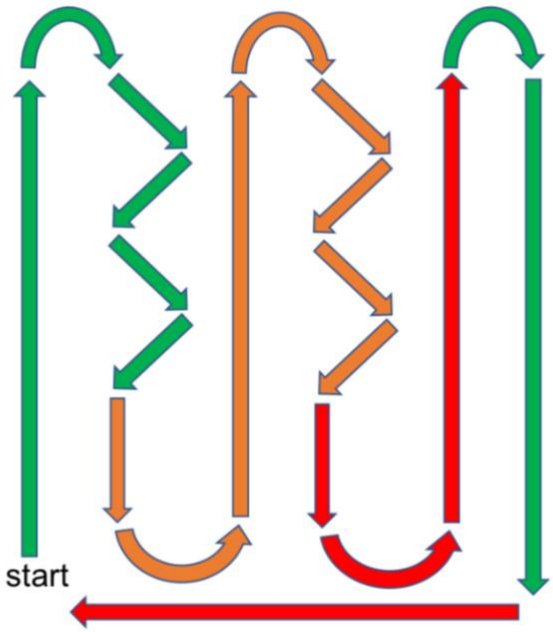


Figure 6. *Schematic of the circuit; green indicates walking, orange indicates jogging, red indicates maximal speed and acceleration*

2.6.2. Maximal Sprints

A minimum of two trials of maximal sprints over 40 m for each player (20 sprints total) will be captured. Each sprint will commence outside the VICON capture space and concluding with a rapid deceleration inside the capture space.

2.6.3. 2v2 Small-Sided Games

Two 2v2 small-sided games will be played in the VICON capture space. During each game players will attempt to maintain ball possession without the possibility of scoring. The game will have a minimum duration of 2 minutes each.

2.6.4. 5v5 Small-Sided Game

One 5v5 small-sided game will be played in the VICON capture space. During this game players will attempt to maintain ball possession without the possibility of scoring. The game will have a minimum duration of 3 minutes.

2.7. 2D & 3D Ball Tracking

Providers participating in ball tracking analysis must ensure continuous tracking and data submission from the start to the end of each session capture, as indicated by the whistle (Section 2.5). In addition to small-sided games, dedicated ball capture sessions will be conducted as outlined below:

2.7.1. 10 m, 20 m, and >20 m Passes

Ball and Kick metrics will be measured from three players performing 20 kicks each (10 x left foot, 10 x right foot) for each passing distance over approximately 10 m, 20 m, and >20 m within the 30 x 40 m VICON capture space. Players will be instructed to receive and kick the ball with the same foot for each kick (i.e. stop ball with right foot, kick with ball with right foot). This drill will be used for ball tracking and kick detection metrics.

2.7.2. Ball Dribble – Continuous Tracking

Ball position will be measured during a continuous dribbling task in which one player will dribble the ball around the inner perimeter of the VICON space. The ball will be dribbled once in the clockwise and once in the anticlockwise directions around the space.

2.7.3. Throw-ins

Ball position during throw-ins will be measured from two players performing 20 throws, with 10 x throws at a ball throw height of approximately 2 m, and 10 x throws at a height of approximately 4 m. The ball will be allowed to bounce twice (see Figure 7). Trajectories from VICON and the provider will be compared in the X, Y, and Z axes. For higher throws that exceed the VICON capture volume, analysis will be limited to the recorded data. Maximum height will also be estimated using VICON-derived velocity and projectile motion equations.

The trajectory between the first and second bounce will be fitted with a quadratic curve, and the R^2 value will assess the fit quality.

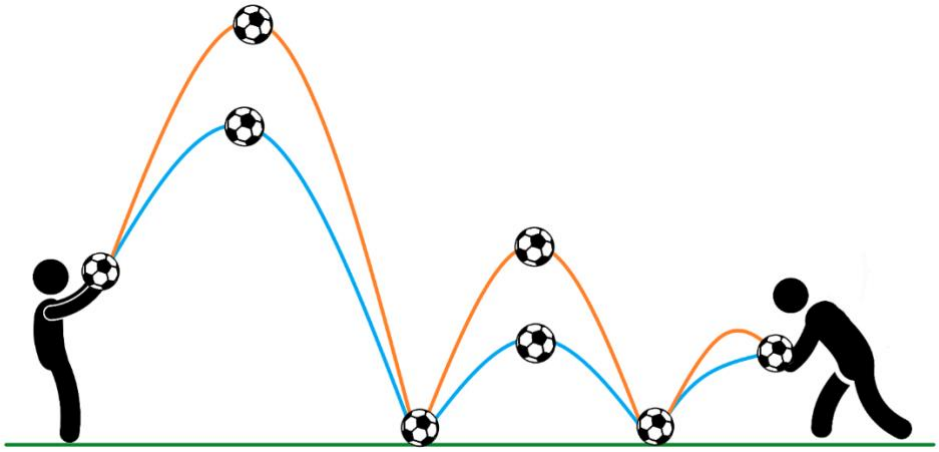


Figure 7. *Throw-ins protocol for 2 m (blue) and 4 m (orange) ball heights*

2.7.4. Small-Sided Games - Continuous Ball Tracking

Ball position (XY) will be measured continuously during the 2v2 and 5v5 small-sided games (see Sections 2.6.3 & 2.6.4) and compared with provider data.

2.7.5. Full Pitch Stationary Position

Ball position accuracy will be assessed using 2-D (XY) coordinates at 10–20 known pitch locations measured by the Total Station (see Figure 8). A test institute member will dribble the ball to each spot, pause it without foot contact for 3 seconds, then move to the next position.

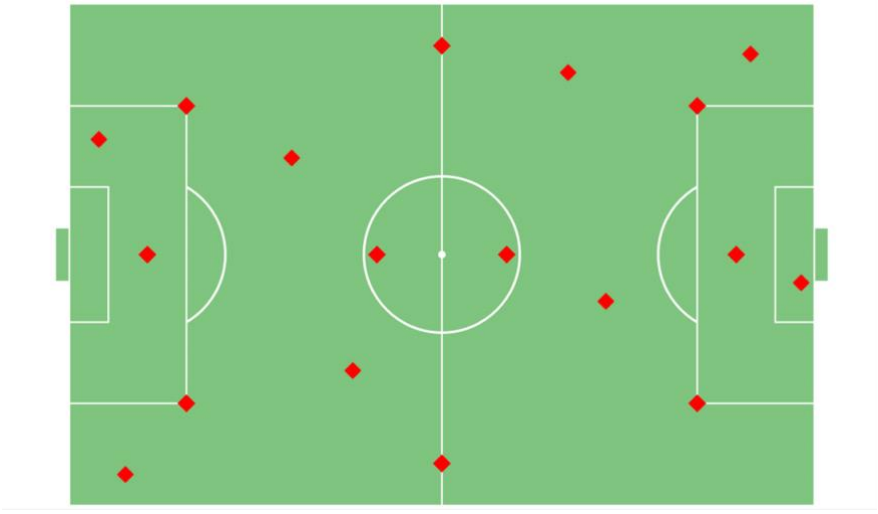


Figure 8*. Example of known positions measured by the total station across the pitch area, these positions are not indicative of exact positions used during test events*

2.8. Live Tracking Data

Applications for live tracking data have increased dramatically over recent years making it important to assess both the quality and the latency of the data. Currently, FIFA considers live data to be data that can be used in near real time which allows for a variety of different latencies depending on the specific use case. Live testing, and certification under the FIFA Quality Programme for EPTS, is currently only possible via a Server Upload Assessment, as explained below:

2.8.1. Server Upload Assessment

The objective of this test is to determine the latency of a data set. In this context, latency refers to the difference in time between when the data is generated and when it is available for use.

This test method uses a gRPC API provided by FIFA. Providers are required to transmit their live tracking data through this API at the same sampling frequency as their system. Each message must be timestamped by the provider with both the collection time (i.e. when the data was generated) and the publication time (i.e., when the message is transmitted via gRPC).

Upon reception, the test institute - acting as the client in this process - records the reception timestamp for each message and forwards the received data to permanent storage within Google Cloud Platform.

The publication latency of the data is calculated by subtracting the collection timestamp from the publication timestamp, both provided by the system under test. The reception latency is calculated by subtracting the collection timestamp from the reception timestamp (generated by the test institute). Additionally, the transmission latency introduced by the gRPC process is assessed by subtracting the publication timestamp (generated by the provider) from the reception timestamp (generated by test institute).

The same data set is accessed by the test institute, in order to analyse the velocity and positional accuracy as per the EPTS Test Manual. For this reason, providers only need to upload one live data set via gRPC in order to have both the latency and accuracy of the data set determined.

For further details about the Server Upload Assessment, including data structure and transmission specifications for gRPC-based submissions, a separate ‘Guidelines: EPTS Live Tracking Data Assessment via gRPC’ document is available in Appendix C.

3. Data Analysis

3.1. Provider Data Submission

Provider data MUST be provided in .csv format. The file MUST be named using the following naming convention:

'systemname_playerid_DDMMYYYY_time'.csv

Where ***'systemname'*** is the name of the provider system being tested, ***'playerid'*** is the unique player jersey number (i.e., black19), and '***DDMMYYYY_time***' is the date of test as Day Month Year and start time of the test session.

Each file provided will be checked for consistency of sample rate (i.e., dropped data points, inconsistent sample rate) and will be resampled at 50 Hz using linear interpolation to allow synchronisation with the VICON data. Further to the raw data provided by providers, a 2nd order Low Pass Butterworth Filter with a 1 Hz cut-off will be applied to all providers' velocity data. Ball tracking data will be assessed at the sample rate given by the provider.

Specific provider data submission requirements for each metric type are provided in Sections 3.2.1 to 3.2.4 below.

3.1.1. Position & Speed

For providers participating in speed and position metric evaluation, all files MUST include the following data columns:

i. Frame
ii. Time (if applicable)
iii. X (or latitude)
iv. Y (or longitude)
v. Speed

* Unless otherwise agreed with the test institute directly prior to testing, GNSS positional tracking data will not be analysed.

3.1.2. Acceleration

For providers participating in acceleration metric evaluation, the test institute requests the following data submissions:

i. Filtering applied to speed data prior to the calculation of acceleration

ii. The calculation used to quantify acceleration

iii. Information on whether the vector components of acceleration – specifically along the X and Y axes – are maintained

iv. Filtering applied to the acceleration data

v. Details on how a positive and negative acceleration (i.e. deceleration) event is categorized. For example, what thresholds are implemented, what duration of time is required to register an 'event'

3.1.3. Load

For providers wishing to participate in load metric evaluation, the test institute needs the following data submissions:

i. Components of the Inertial Measurement Unit

ii. Sampling rate of individual components

iii. Components that contribute to the quantification of 'Load' and associated calculations

iv. Providers using custom or proprietary load calculation (e.g. PlayerLoad™) are encouraged to clearly document and communicate their method and its components to the test institute prior to testing

3.1.4. Ball & Kick Metrics

For providers wishing to participate in ball and kick metric evaluation, the test institute requests the following data submissions:

i. Timestamp of detected kick with associated speed

ii. Player ID and, if possible, detected foot

iii. Optional (preferred): Timeseries of ball and/or foot speed in a fixed time window around each event (e.g. 2 s)

3.2. Measurement Units

Measurement units used to process and evaluate provider data are listed in Table 1. If data is submitted in alternative measurement units (e.g. km/h, cm), it will be converted to match those in Table 1. However, providers using different measurement methods (e.g. load index at the foot vs. pelvis) should inform the test institute prior to their test session.

Table 1. *Metrics evaluated under the FIFA EPTS Quality Programme, including new metrics added from 2025.*

Type	Metric	Unit of Measurement
Speed	Speed	m/s
Position	2-D Position	m
NEW METRICS FROM 2025*		
Acceleration	Acceleration	m/s^2
	Acceleration Efforts	Count (n)
Load Index	Acceleration Index	m/s^2
	Load Index	Arbitrary Unit (AU)
Technical	Ball Contact	Count (n)
	Ball Contact Time	s
	Kick Detection	Count (n)
	Kick Detection Time	s
	Foot-to-Ball Contact Time	s
	Foot Peak Speed	m/s
	Foot Mean Speed	m/s
	Differences between Left & Right Foot	Count (n), m/s, s
	2-D Ball Position	m
	2-D Ball Speed	m/s

3.3. 3-D Motion Capture Data Preparation

Motion capture data will be provided as individual drill files for multiple players, sampled at 100 Hz, with files split by player and drill. Each file includes X, Y and Z coordinates (in mm), including speed for all axes. Absolute speed will be calculated from the X and Y speed components. Criterion speed will be calculated from VICON COM speed data using the three-point central difference method. To enable comparison with 10 Hz and 25 Hz provider data, the X, Y, and velocity data will be linearly interpolated to 50 Hz,smoothed with a 5-point moving average, and filtered using a 2nd-order low-pass Butterworth filter with a 1 Hz cut-off.

Ball detection observations will include an XY ground position and a timestamp. Ball tracking data will be smoothed using a 5 Hz Butterworth recursive digital filter, with the cut-off determined through wavelet analysis, residual analysis, and visual inspection of peaks and troughs.

3.4. Data Synchronisation

Individual player and drill files (VICON) will be imported and synchronized with the corresponding provider player data. To eliminate filtering phase shift effects, speed will be recalculated from the provider's X and Y coordinates using the same method applied to the VICON data (Section 3.3): a 3-point central difference followed by a 2nd order low-pass Butterworth filter with a 1 Hz cut-off.

Initial synchronization will be performed using cross-correlation between the recalculated provider speed and the VICON speed to determine the optimal shift that maximises correlation. The data will then be trimmed and combined. Further synchronization of the provider's speed trace will involve shifting it ±50 data points at single-point intervals, calculating the Root Mean Squared Difference (RMSD) at each step (see Section 3.6 for RMSD methods). The shift with the lowest RMSD will be selected for analysis. Finally, the X and Y coordinates from the provider and VICON systems will be time-aligned for analysis. The provider's original speed trace will also be independently shifted to optimise alignment. The final dataset for statistical analysis will include both VICON and provider data.

3.5. Position Alignment

To examine the difference in player and ball position data, X and Y co-ordinates shall be rotated to match the X and Y co-ordinates from VICON data. This will be achieved by rotating the provider's data by 1 degree through 360 degrees until the lowest error in position data is achieved. Once the closest 1-degree rotation is known, the provider's X and Y co-ordinates will be further adjusted to the closest 1/100 of 1 degree either side of the best alignment with the VICON data.

3.6. Statistical Analysis

Alignment between provider and VICON data for both position (in m) and velocity ($m.s^{-1}$) will be assessed using both absolute and relative measures. For absolute measures, the mean difference between the two data sources will be obtained to provide an indication of any systematic differences between the two data sources. Values will be presented at an aggregated level (all data combined from the player trials). Specific statistical methods that will be used for evaluation of given metrics are outlined below.

3.6.1. Root Mean Square Difference

Root-mean square difference (RMSD) is used to determine the level of agreement between timeseries data (e.g. timeseries acceleration, load index; foot velocity), and is calculated as the square root of the mean squared differences between the reference and provider datasets:

$$RMSD = \sqrt{\frac{\sum(Reference_i - Provider_i)^2}{n}}$$

3.6.2. Mean Absolute Difference

Mean Absolute Difference (MAD) will be used to determine the level of agreement between systems position data and the remaining kick metrics (e.g., foot-to-ball contact time; peak foot speed), and measures the mean of the absolute differences between the reference and provider datasets:

$$MAD = \frac{\sum|Reference_i - Provider_i|}{n}$$

3.6.3. Sensitivity

Sensitivity (or True Positive Rate) will be used to assess a provider's ability to accurately detect kicks. Sensitivity considers True Positive (those successfully detected by both the provider and the reference system) and False Negative kicks (kicks detected by the reference system, but not by the provider):

$$Sensitivity = \frac{TP}{TP + FN}$$

4. Position & Speed - Criteria for Product Accreditation

The product accreditation reports granted to providers after a test event and published on the FIFA Quality Programme website are regularly reviewed to improve outcomes for the end user. A sample of this report is included in Appendix A. Specific analysis methods for these reports are provided below.

4.1. Position

The difference between the provider X and Y coordinates and VICON X and Y coordinates shall be quantified as the straight-line difference between the coordinates.

4.2. Speed

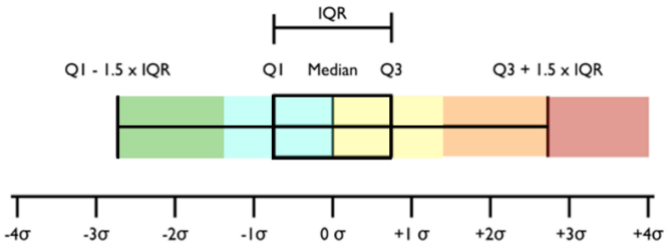
Speed can be defined as how fast a player or ball is moving over time and is defined by the equation below.

$$\boldsymbol{Speed} = \frac{\boldsymbol{Distance}}{\boldsymbol{Time}}$$

The quantified difference between the provider and VICON speed shall be used to evaluate and rate provider systems.

4.3. Rating System

Providers will be evaluated for speed and position (where applicable) using a five-level scale with colour coding, rather than specifying exact velocity or position differences. Speed and position differences will be rated within FIFA velocity bands (0-5 km/h, 5-10 km/h, 10-15 km/h, 15-20 km/h, 20-25 km/h, +25 km/h). To determine the rating system, the analysis conducted on >1,000,000 data points from over 30 systems at the past three EPTS test events form the basis of the ranking system, these are reviewed on a yearly basis. Box plot values have been used to develop thresholds to inform this grading based on current industry standards; these are as follows:



Legend

Well-above	Above	Standard	Below	Well-below

5. New Metrics

The new metrics evaluation methods that will be available for providers for quality evaluation from 2025 are detailed below. Please note these metrics will not form part of the provider certification process in 2025, however providers are given the option to be evaluated on these metrics. New metric evaluation results are intended to be either included in the published test reports or reserved for internal use by each provider. Examples of potential report figures for new metrics are provided in Appendix B.

5.1. Acceleration Evaluation

Acceleration can be defined as the rate of change in speed as per the equation below.

$$\boldsymbol{Acceleration} = \frac{\boldsymbol{Change\ in\ Speed}}{\boldsymbol{Change\ in\ Time}}$$

Criterion acceleration will be calculated from VICON COM speed data using the three-point central difference method. Acceleration will either be supplied by the provider or derived from their speed data. Different acceleration calculation methods for providers based on technology type are outlined in Table 2. These methods are based on how each tracking technology derives acceleration from its data.

Table 2. *Proposed acceleration calculation method for different technology types.*

Technology Type	Calculation Methods
Optical & LPS systems	Derived as the second derivative of position
GNSS-based systems	Derived as the first derivative of speed via Doppler shift
IMU-based systems	Directly measured via accelerometers (can include magnetometer and gyroscope corrections)

Player and ball tracking data often contains a degree of noise that must be treated to accurately quantify movement patterns. As such, filtering of the acceleration data using a low-pass filter designed to eliminate high-frequency noise is employed based on the residual analysis technique described by Winter [1] will be conducted to control for this noise. This method systematically identifies the optimal cut-off frequency by assessing the degree of signal distortion across a range of potential values. To inform this, a series of low-pass filters were applied to legacy EPTS data, spanning from 0.1 Hz to 10 Hz in 0.1 Hz increments. At each step, the RMSD is calculated. These values are then plotted to identify the point at which the RMSE met the noise threshold. This point marks the optimal cutoff threshold where a lower frequency would begin to distort the true signal of a device, and a higher frequency would fail to sufficiently reduce noise. For position, speed and acceleration-based metrics, it was established that a 1 Hz cut off was optimal cut-off frequency.

5.1.1. Time-series Acceleration

i. Time-series Acceleration Definition

Time-series acceleration refers to the rate of change of speed over time, typically derived from sequential position or speed data points.

ii. Assessment Methods

To ensure optimal alignment with VICON data, acceleration data from the tracking system will be shifted in time using the data synchronisation techniques described in Section 3.5. Time-series acceleration accuracy will be quantified by calculating the RMSD relative to VICON acceleration. Accuracy will also be analysed across the acceleration spectrum to evaluate performance at different intensities

5.1.2. Acceleration Efforts

i. Acceleration Efforts Definition

Acceleration efforts are defined based on specific criteria, including a minimum acceleration threshold and/or duration.

ii. Assessment Methods – VICON v Provider

Acceleration efforts will be defined depending on the following effort definition criteria or as specified by the providers (Table 3):

Table 3. *Summary of acceleration effort thresholds. Adapted from Ellens et al. [2].*

Threshold Type	Acceleration Value (m/s^2)	Mandatory or Optional	Notes
Low Threshold	> 1.0	Optional	May be used to detect low-intensity accelerations
Moderate Threshold	> 2.0	**Mandatory**	Standard reference threshold used for evaluation and comparison
High Threshold	> 3.0	Optional	Useful for distinguishing explosive actions
Provider-Defined	Custom (e.g., >2.5)	Optional	Must be clearly documented and justified by provider before testing

Acceleration effort accuracy will be quantified via a sensitivity and specificity analysis. Sensitivity assessment of how many acceleration efforts were correctly detected will be provided based on the True Positive (acceleration

effort detected by the technology provider and VICON system) and False Negative (acceleration effort by provider but detected by VICON) acceleration effort detections by the provider. Specificity will be measured to the extent to which the system avoids False Positive (acceleration effort detected by provider, but not VICON) acceleration effort detections. Furthermore, Mean Absolute Percentage Error (MAPE) will be used to provide a percentage error of provider acceleration efforts compared to VICON.

5.2. Load Evaluation

5.2.1. Load Index

i. Load Index Definition

The difference between the quantification of acceleration data collected by a provider's device when compared to a criterion measure of acceleration data.

ii. Assessment Methods

To assess the accuracy of load index outputs from IMU devices, data from Blue Trident Inertial Measurement Units (IMUs) sampling at 1600 Hz will be used as the criterion measure. These IMUs can be synchronised with the VICON motion capture system used during data collection. Data will be collected continuously throughout the test session, with the start time of each drill manually recorded.

Acceleration data from each individual axis (X, Y, Z) will be band-pass filtered using a 2nd-order Butterworth filter with a frequency range of 0.5 – 20 Hz. Load Index will be calculated using the vector magnitude formula or a minor variation, dependent on information given by the provider:

$$\boldsymbol{Load\ Index} = \sqrt{\boldsymbol{X^2 + Y^2 + Z^2}}$$

Blue Trident data will be up-sampled to 2000 Hz and then down-sampled to 100 Hz to match the processing requirements. The provider's IMU data will also undergo filtering and summation across the three axes to establish a consistent 100 Hz sample rate. Data will be converted to $m \cdot s^{-2}$ or arbitrary units (AU) depending on provider data submission.

To align the Blue Trident data with the provider data, a sliding window method will be applied. The Blue Trident signal will be shifted incrementally across the provider's signal. At each time step (1/100th of a second), the RMSD of the Load Index will be computed between the two systems. Data will be trimmed to match the start and end times of each drill. The synchronization accuracy for each drill and player will be visually inspected by plotting both signals for comparison.

5.3. Ball & Kick Metric Evaluation

VICON speed data for relevant lower body landmarks will be determined using the three-point central difference method. Specific VICON markers used will be heel, ankle, toe, or foot centre (derived by averaging the X, Y, Z position of heel, ankle and toe markers). VICON markers used will be determined as the closest aligned to the providers tracking segment. VICON kick events will be identified and time-stamped using the kicking phases defined below [3] and shown in Figure 9.

- *Kicking foot toe off*: foot leaves ground to initiate backswing of kicking leg.
- *Forward swing phase:* frame from which kicking foot reaches its maximum altitude behind the players body through to the frame in which the foot contacts the ball.
- *Ball contact:* the frame when the foot first contacts the ball and the ball starts its trajectory.
- *Ball release:* the frame where the foot is no longer in contact with the ball.
- *Follow through phase:* the frame of ball release through to the maximum altitude of the limb in front of the player's body.

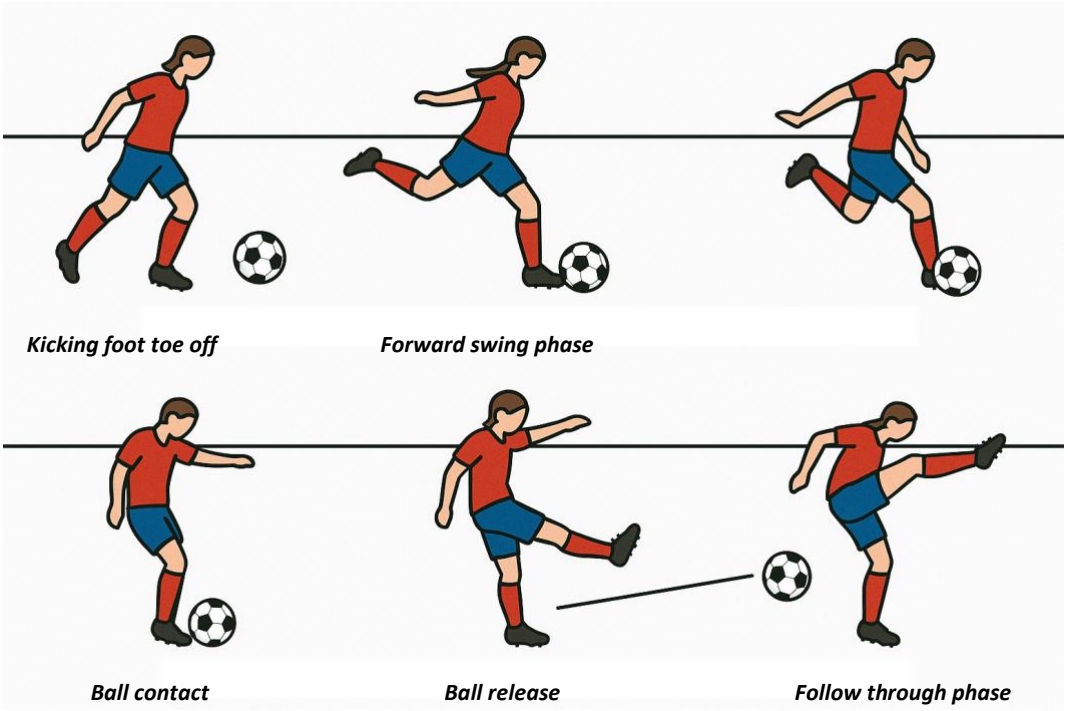


Figure 9. *Visual representation of specific kick phases adapted from Ángel-López, Segura-Giraldo [3]*

The ball centre tracking provides spatiotemporal tracking of movements within the VICON capture space. Ball marker data will be collected via VICON Nexus. The comparison of the above VICON kick event data to specific provider ball metrics will be conducted using the evaluation methods below.

5.3.1. Ball Contacts & Kick Detection

i. Kick Detection Definition

A kick detection event will be defined as any kick event that includes all kick phases listed above and will require the transference of ball possession from one player to another.

ii. *Ball Contact Definition*

Ball contacts will be defined as the frame from which foot contact is made with the ball.

iii. Kick Event Time Definition

Time reference point in which initial foot-to-ball contact is made for either kick detection or ball contact assessments.

iv. Assessment Methods

To access kick detection accuracy (including sensitivity and specificity), kick events recorded in VICON and coded from video footage will be compared against providers kick event outputs. Alignment of VICON and coded video footage with provider data will be conducted using kick time synchronization for each trial (see Section 3.4). Kicks recorded by the provider occurring outside the VICON capture space will not be included. Sensitivity assessment of how many actual kicks were correctly detected will be provided based on the True Positive (kick detected by the technology provider and VICON system) and False Negative (kick missed by provider but detected by VICON) kick detections by the provider. Specificity will be measured as how well the system avoids False Positive (kick detected by provider, but not VICON) kick detections. Furthermore, MAD will be used to assess the accuracy of the time difference between detected kicks and VICON kicks detected.

5.3.2. Foot-to-Ball Contact Time

i. Contact Time Definition

Time in seconds between initial foot contact and ball release.

ii. Assessment Methods

To assess foot-to-ball contact time accuracy, foot-to-ball contact events recorded in VICON and coded from video footage will be compared against the provider's foot-to-ball contact event outputs.

Foot-to-ball contact time will be defined for VICON as the ball contact and ball release phases (previously defined). A foot-to-ball contact event will be classified as any event that includes both phases. Alignment of VICON kick phases with provider data will be conducted using time synchronisation for each trial (described in Section 3.4).

Foot-to-ball contact events recorded by the provider occurring outside the VICON capture space will not be included. MAD in seconds will be used to assess the accuracy of the time difference between detected foot-to-ball contact events and those recorded by VICON.

5.3.3. Foot Speed – Peak Speed

For accuracy assessment of foot peak speed, a provider must also provider kick detection outputs (see previous methods).

i. Peak Foot Speed Definition

Maximum speed recorded between forward swing phase and end of follow through phase.

ii. Assessment Methods

To assess peak foot speed accuracy, kick peak foot speed recorded in VICON will be compared against the provider's peak foot speed outputs. A peak foot speed event will be classified as the frame in which this maximum velocity occurs. Alignment of VICON foot speed data with provider data will be conducted using the time synchronization methods described in Section 3.4. Peak foot velocity events recorded by the provider occurring outside the VICON capture space will not be included. MAD will be used to assess the accuracy of the peak foot speed difference detected between the providers outputs and foot speed events recorded by VICON.

5.3.4. Foot Speed – Mean Foot Speed

For accuracy assessment of mean foot speed, a provider must also provider kick detection outputs (see previous methods).

i. Mean Foot Speed Definition

Mean speed recorded between the forward swing phase and the end of the follow-through phase.

ii. Assessment Methods

Mean foot speed accuracy will be evaluated by comparing foot speed recorded in VICON with the provider's foot speed outputs over a fixed time window around each kick event (e.g. 2s). VICON foot markers — heel, ankle, toe, or foot centre (averaged from heel, ankle, and toe markers) — will be selected based on their closest alignment to the provider's tracking segment.

To synchronise VICON and provider kick speed data, a sliding window method will be applied. The VICON speed signal will be incrementally shifted across the provider's signal, computing the RMSD at each time step (1/100th of a second). Data will be trimmed to match the start and end times of each kick and exported. Synchronisation accuracy will be visually inspected by plotting both signals for comparison.

Mean foot speed accuracy will be quantified using RMSD between provider outputs and VICON foot speed. Correlations between provider-reported mean foot speed per kick and VICON mean foot speed will also be analysed and visualised.

5.3.5. 2-D Ball Position & Speed

i. 2-D Ball Position Definition

The 2-D position (X and Y coordinates) of the ball centre captured within the VICON capture space.

ii. 2-D Ball Speed Definition

The 2-D speed of the ball centre captured within the VICON capture space. For VICON, ball speed is derived from the X and Y coordinate positions using the three-point central difference method. If ball speed data is provided by the provider system, it will be used directly. If ball speed data is not supplied by the tracking system, it will be derived using the same method as VICON.

iii. Peak Ball Speed Definition

Absolute peak ball speed is the maximum speed output recorded for each ball tracking trial. For kick-specific data, the peak ball speed will be calculated for each kick, with the peak speed of the ball being reported as the maximum speed output recorded from the point of ball release from one player to initial foot contact of another player.

iv. Mean Ball Speed Definition

Mean ball speed is the average speed output recorded for each ball tracking trial. For kick data, the mean ball speed will be calculated for each kick, with the mean speed of the ball being measured from the point of ball release from one player to initial foot contact of another player.

v. Assessment Methods

The continuous XY position of the ball will be determined using the VICON Tracker capture system and ball model (see Section 2.4.4). Tracking will be conducted for the following tasks: 2v2, 5v5, continuous dribble, kick trials (short, medium, and long kicks), and throw-ins (including ball bounces). Where necessary, VICON Tracker data will be interpolated using an interpolation function, with a maximum window of 5–100 frames depending on the data section. Providers participating in ball tracking analysis must track the ball continuously, ensuring data submission from the start to the end of capture.

VICON Tracker ball data will be provided as individual drill files and will be processed according to the methods defined in Section 3.3. Files will then be time synchronised using ball speed following the methods detailed in

Section 3.4. For 2-D ball position data, provider XY coordinates will be rotated and translated to align with VICON Tracker data using methods previously defined in Section 3.5.

Once aligned and synchronised, data will be compared to assess relationships between position and speed traces from VICON Tracker and the provider. Specifically, RMSD will be used to determine the level of agreement between speed metrics (e.g., 2-D Ball Speed, Peak Ball Speed, and Mean Ball Speed). The level of agreement between 2-D Ball Position will be measured using MAD between the VICON system and provider datasets

6. References

1. Winter, D.A., Biomechanics and motor control of human movement. 2009: John wiley & sons.
2. Ellens, S., et al., Techniques to derive and clean acceleration and deceleration data of athlete tracking technologies in team sports: A scoping review. Journal of Sports Sciences, 2022. 40(16): p. 1772-1800.
3. Ángel-López, J.P., et al. Kinematic soccer kick analysis using a motion capture system. in VII Latin American Congress on Biomedical Engineering CLAIB 2016, Bucaramanga, Santander, Colombia, October 26th-28th, 2016. 2017. Springer.

Appendix A: Example Product Certification Report (2024 Version)



FIFA EPTS TEST REPORT

EPTS Performance Test

Product Name	Product name
Partner	Partner name
EPTS Product Type	Optical System
Quality Level	FIFA Quality
Data Submission	Player - Centre of mass Post data submission
Test Date	Choose an item.
Certification Date	Choose an item.
Submission Data Type	Post-Match Performance Data

FIFA

INTRODUCTION

Introduction
The EPTS Performance test offers a four-year FIFA Quality certification for tracking providers who generate player (either centre of mass, skeletal or both) and, in certain cases, ball tracking data. This process is carried out by quantifying the accuracies of Global Navigation Satellite Systems (GNSS), Local Positioning Systems (LPS) and Optical Tracking Systems (OTS). The certification is offered to providers who publish the accuracy of their systems. The test, conducted at an independent stadium, consists of the players and the ball being fitted with reflective markers before performing football specific drills. During these drills Vicon motion capture and technology provider data is collected simultaneously. Post-event, the data is temporarily and spatially synchronised before the differences between the data sets are calculated and visualised in the below report.

Declaration of conformity	
This report details the results of EPTS Performance Tests carried out in accordance with the 'Handbook of Test Methods for EPTS Devices'. For further information about the exact test procedure, please refer to the 2022 edition of the Handbook of Test Methods for EPTS devices.	
Technician	Jade Haycraft
Date	30.07.2024
Signature	
Report checked by	Sam Robertson
Date	30.07.2024
Signature	

FIFA

PLAYER TRACKING TEST CONDITIONS

Test details	Centre of mass - player data
Test Date	Click or tap here to enter text.
Test Location	Municipal Stadium of Leiria, Dr. Magalhães Pessoa
Test Institute	Victoria University
Test Number	Click or tap here to enter text.
Test Manual	EPTS Performance Standard | 2022
Number of VICON Cameras	40
Type of VICON cameras	35 x Valkyrie, 5 x Vantage V16
Player Participant Details (number & sex)	5 Male and 5 Female
Approximate Player Age	18



FIFA

SYSTEM DETAILS

System Details	Centre of Mass Player Data
Test Number	Click or tap here to enter text.
Test Manual	EPTS Performance Standard | 2022
EPTS Product Type	Optical System
Assessment Metrics	Post-Match Performance Data
Number of Provider Cameras	Click or tap here to enter text.
Provider Camera Model Names	Click or tap here to enter text.
Submission Time	< 1 hour
Median Latency (if applicable)	Click or tap here to enter text.

Fulfilment of test requirements

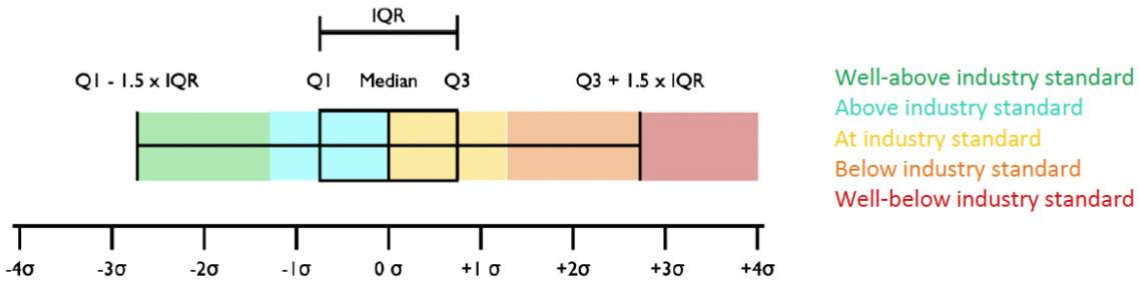
Test block	Capture & submission
Circuit	Yes
2 v 2 Game	Yes
5 v 5 Game	Yes
Sprints	Yes
Full Pitch Coverage	Yes

Analysis interpretation

Measure	Definition
Root Mean Square Difference (RMSD)	A commonly used measure of accuracy based on the standard deviation (σ) of the differences between the manufacturer and Vicon system.
Data Points Compared with Vicon	Varies depending on the quality of Vicon capture, as only the highest quality data is used for comparative purposes.
Live Data Submissions	Live data is considered as data that can be used in real time; this varies depending on the application.

FIFA

Rating System (including legend)



PLAYER TRACKING RESULTS – CENTRE OF MASS

FIGURE 1: Rating by FIFA Velocity Band

Rating by Velocity Band

	0-5 km/h	5-10 km/h	10-15 km/h	15-20 km/h	20-25 km/h	25+ km/h
Velocity						
Position						

Legend

Well-above	Above	Standard	Below	Well-below

FIFA

System Differences by Speed Band

FIGURE 2: Mean speed difference, expressed as Root Mean Square Difference (RMSD) ± Standard Deviation (SD), across all players by FIFA speed zones.

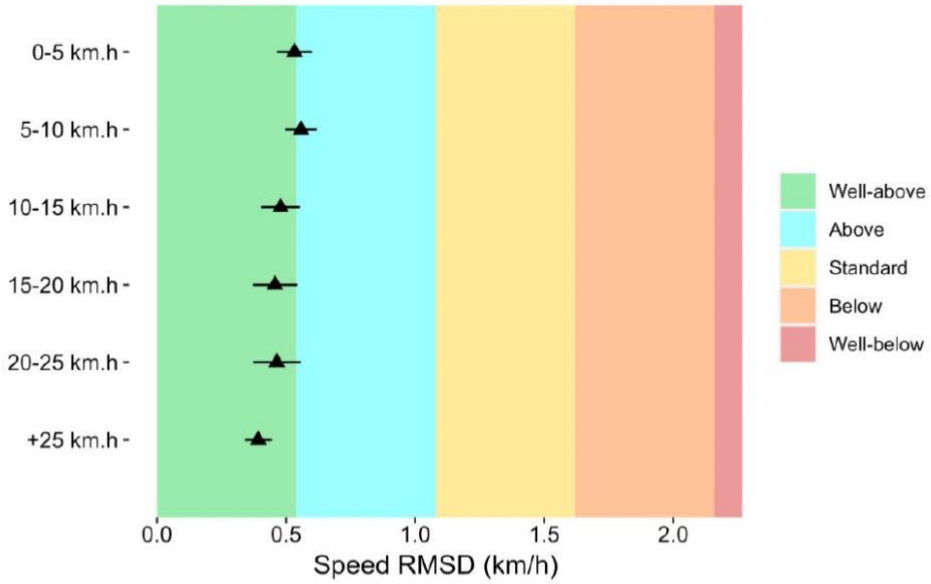
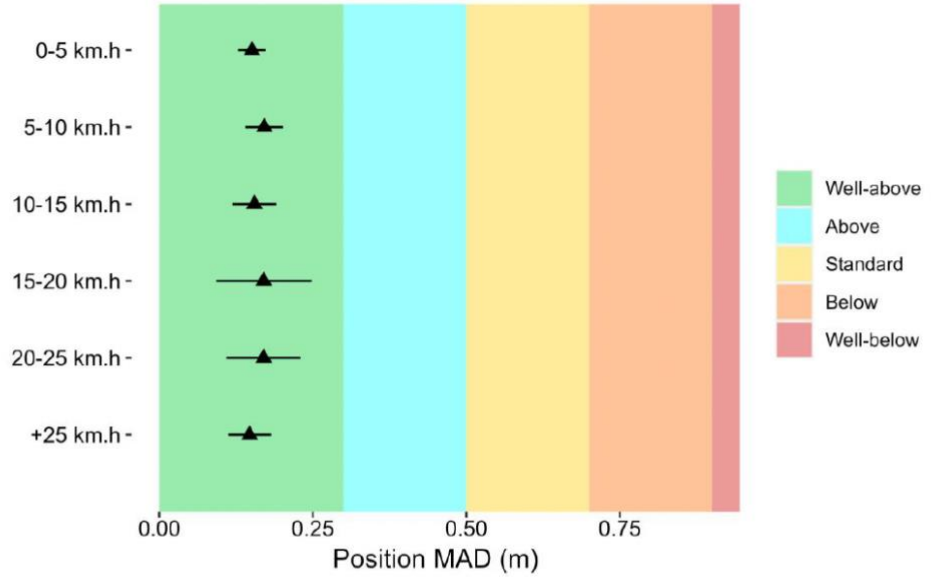


FIGURE 3: Mean position difference, expressed as Mean Absolute Difference (MAD) ± Standard Deviation (SD), across all players by FIFA speed zones.



FIFA

Model of Differences Relative to Criterion

FIGURE 4: Analysis of system speed difference relative to Criterion (VICON) Speed. Upper and Lower boundaries are the mean positive and negative speed differences observed across all manufacturers for the 2023 and 2024 FIFA EPTS test events. The linear equation can be used to systematically account for the differences between systems, with the standard error of the estimate (SEE) of the model also shown.

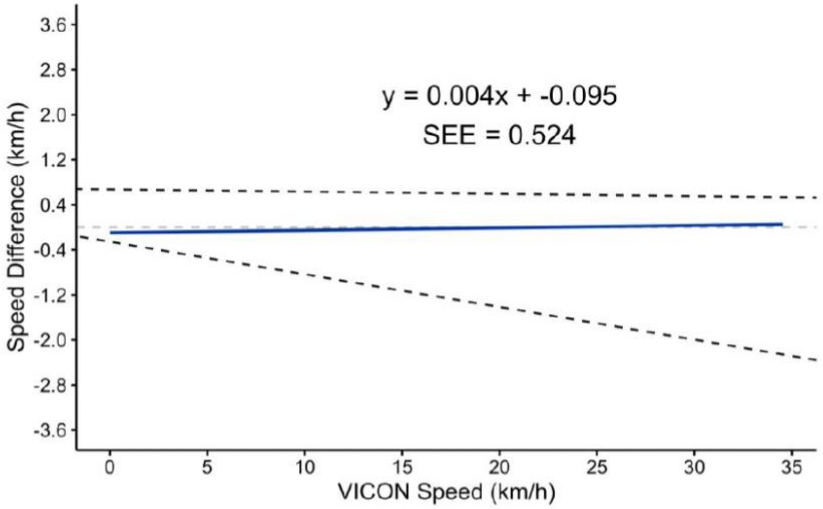
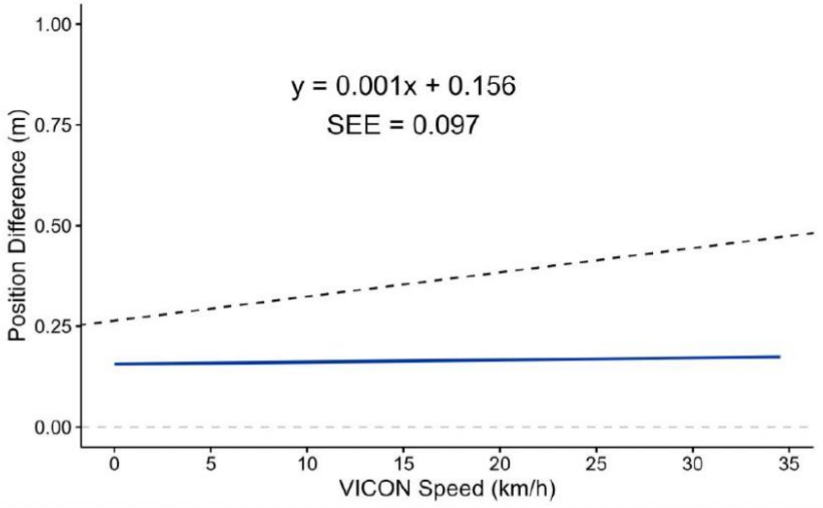


FIGURE 5: Analysis of system position difference relative to Criterion (VICON) Speed. The upper boundary (dotted line) represents the mean position differences observed across all manufacturers for this test event. The equation can be used to systematically account for the differences between systems, with the standard error of the estimate (SEE) of the model also shown. Unlike in Figure 4, no lower boundary is shown due to only absolute differences in position being assessed.



FIFA

Total Distance Comparison & Data Submission Details

FIGURE 6: Comparison between VICON system and provider system of total distance covered by each player, aggregated across all drills. Total distance difference between systems is calculated based only on comparable VICON points.

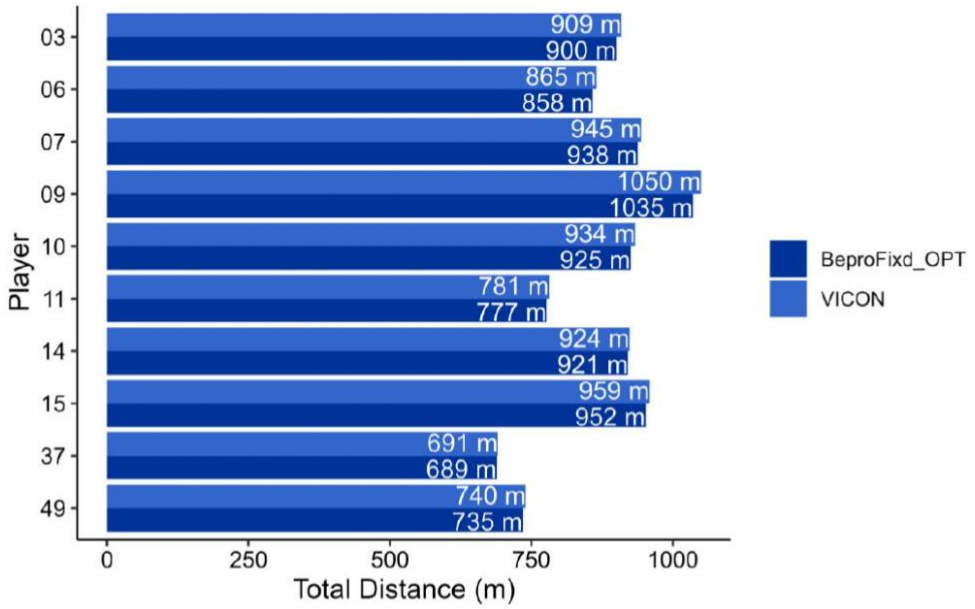


FIGURE 7: Provider data submission completeness (%) by player by drill. Table cells are presented only for players participating in a specific drill. Percentage values relate to the number of provider data points submitted as a proportion of Vicon data points used in the analysis.

	Drill						
Player	**Circuit 1**	**Circuit 2**	**2v2 (a)**	**2v2 (b)**	**5v5**	**Sprint 1**	**Sprint 2**
Player 03	100		100		100	100	
Player 06		100	100		100	100	100
Player 07		100	100		100	100	100
Player 09	100		100		100	100	100
Player 10	100			100	100	100	100
Player 11		100		100	100		100
Player 14		100		100	100	100	100
Player 15	100			100	100	100	100
Player 37	100				100	100	100
Player 49		100			100	100	100

FIFA

Ball Tracking

FIGURE 8: Absolute position difference (m) for the 14 assessed ball positions across the full pitch.

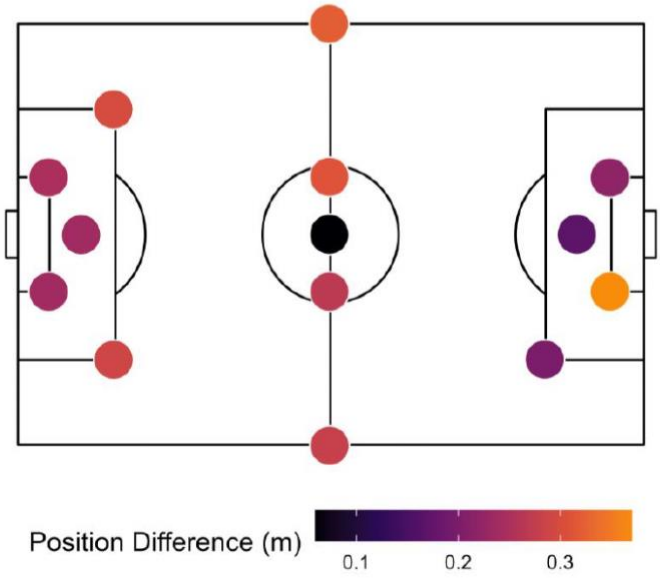


FIGURE 9. Mean speed comparison between systems for the 15 passes completed in the testing

Pass #	VICON (m/s)	Provider (m/s)	Difference (m/s)
1	4.07	4.41	0.34
2	5.32	5.64	0.32
3	4.27	4.43	0.16
4	5.41	5.46	0.05
5	5.96	6.7	0.74
6	6.20	6.72	0.52
7	7.40	7.71	0.31
8	6.86	7.01	0.15
9	6.80	6.79	0.01
10	7.54	7.59	0.05
11	8.50	8.75	0.25
12	8.99	9.25	0.26
13	9.23	9.39	0.16
14	9.20	9.34	0.14
15	9.19	9.45	0.26
Mean	**6.99**	**7.24**	**0.25**

FIFA

FIGURE 10. Mean speed comparison between systems for the 18 passes completed in the testing. Speed calculation based on mean values recorded from foot contact to foot contact of the passing players.

Pass #	VICON (m/s)	Provider (m/s)	Difference (m/s)
1	5.15	5.18	0.03
2	3.65	3.62	0.03
3	5.60	5.50	0.10
4	4.27	4.41	0.14
5	5.69	5.90	0.21
6	7.70	7.62	0.08
7	6.70	6.90	0.20
8	9.88	9.72	0.16
9	7.51	7.66	0.15
10	7.69	7.65	0.04
11	7.98	8.02	0.04
12	10.26	10.39	0.13
13	9.26	9.46	0.20
14	11.60	11.70	0.10
15	10.76	11.10	0.34
16	13.29	13.60	0.31
17	10.04	10.09	0.05
18	8.52	8.60	0.08
Mean	**8.09**	**8.17**	**0.13**

FIGURE 11. Root mean square difference (m) by drill based on continuous data capture

Drill	X-axis	Y-axis	Z-axis	XY	XYZ
Dribble	0.015	0.016	0.047	0.020	0.048
2 v 2	0.037	0.040	0.052	0.051	0.067
5 v 5	0.193	0.201	0.228	0.226	0.259

FIFA

LIVE DATA SUBMISSION

Applications for live tracking data have increased dramatically over recent years making it important to assess both the quality and the latency of the data. At this stage, FIFA considers live data to be data that can be used in real time which allows for a variety of different latencies depending on the specific use case. Live testing, and certification under the FIFA Quality Programme for EPTS, is currently only possible via a Server Upload Assessment as outlined in the **2022 edition** of the EPTS Test Manual.

Server Upload Assessment

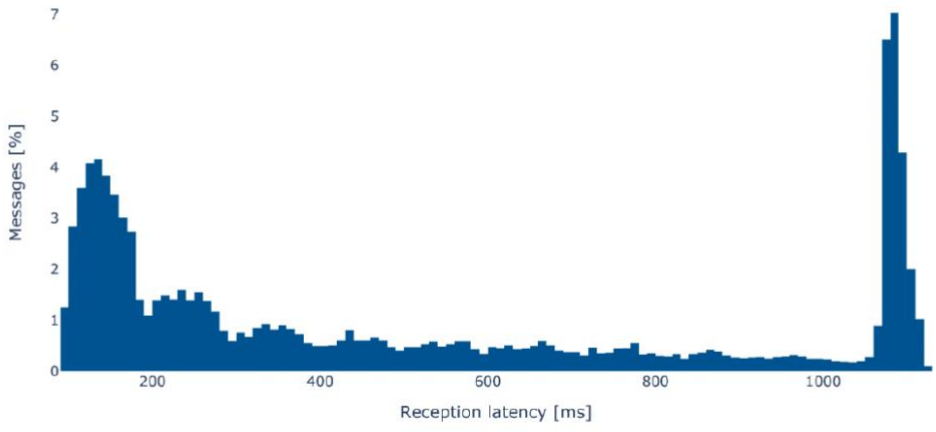
The objective of this test is to determine the latency of a data set. In this test, latency refers to the difference in time between when the data is created and when it is available for use.

This test method uses an open-source message broker software called Rabbit MQ. Tracking providers upload their live data to the platform, Google/MIT then act as a client and publish the messages to the Google Cloud Platform with a timestamp. The latency of how long the data takes from creation to publication time is shown below; this includes the latency caused by the server itself which has been assessed. It is important to note that, whilst RMQ was used to assess latency in this test, a provider may deliver live data to its clients in a different format.

Provider Results

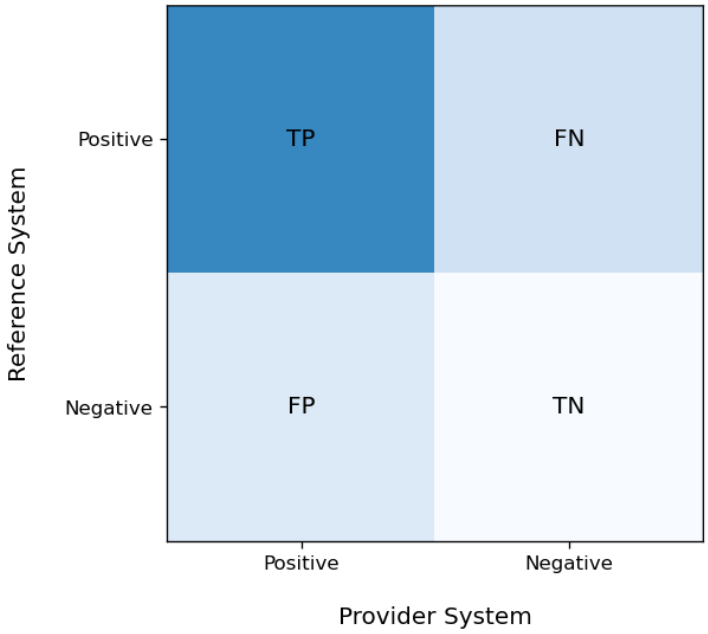
The data below illustrates the variation in upload latency to RMQ:

Data Analysis	Publication	RMQ	Reception
Median latency [ms]	334	36	378
99% latency [ms]	1083	69	1122

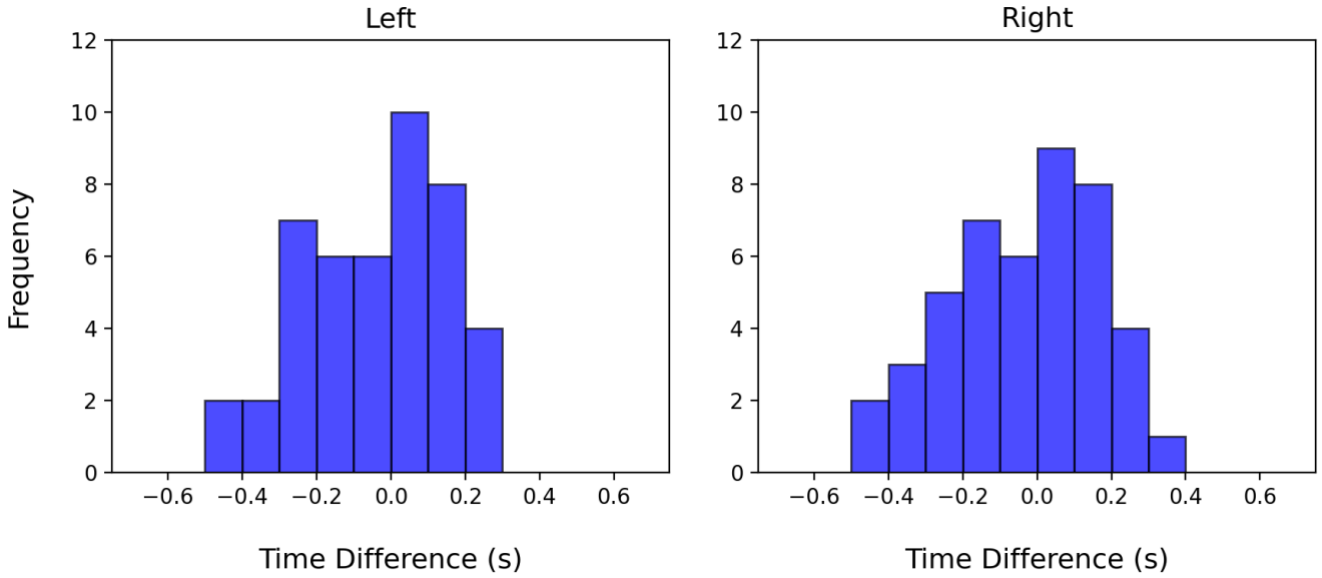


Appendix B: Example Figures for New Metrics

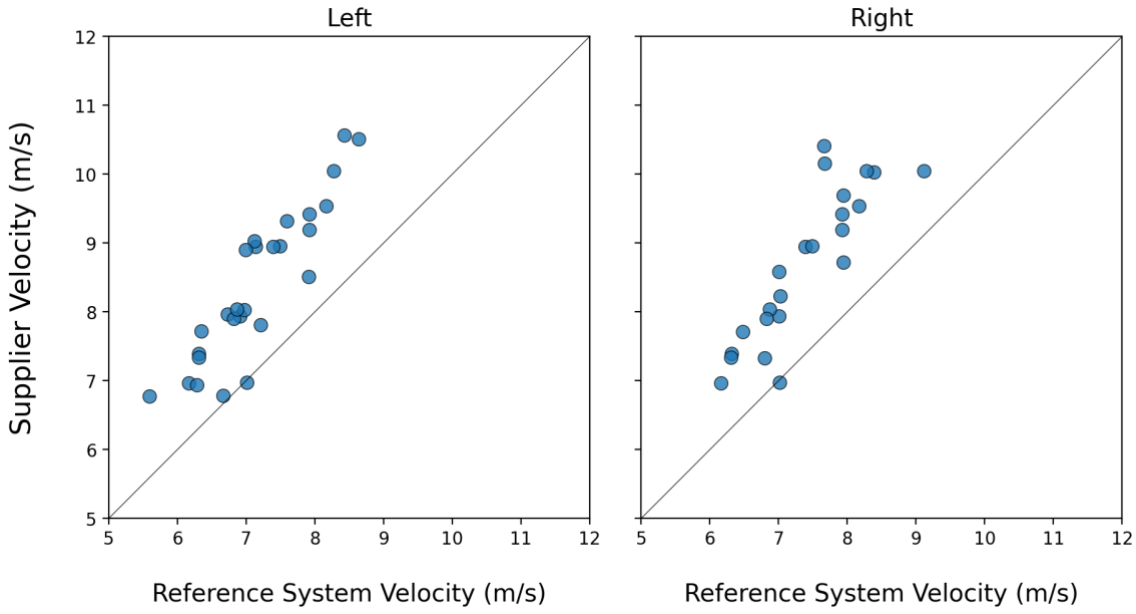
Example visualisation of a provider sensitivity assessment for kick detection.



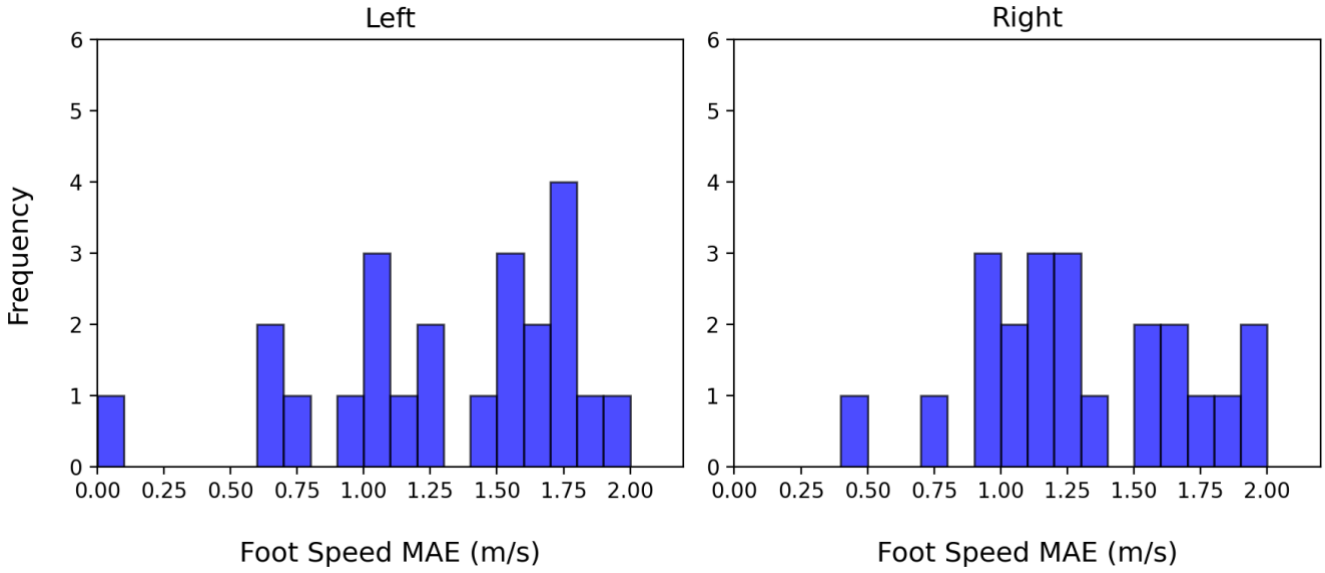
Example visualisation of a kick detection time difference between criterion system and provider system.



Example visualisation of correlation error between criterion system and provider system for ball velocity



Example visualisation of histogram of foot speed error (MAE) between right and left foot kicks.



Appendix C: Guidelines: EPTS Live Tracking Data Assessment via gRPC

FIFA

Guidelines: EPTS Live Tracking Data Assessment via gRPC

I. What is the gRPC?

gRPC is a high-performance, open-source remote procedure call framework. For the purpose of the FIFA Quality Programme for EPTS, it is used to assess the latency of live tracking submitted by various data providers (Provider). The transition from the previous RabbitMQ system to gRPC enhances reliability, scalability and efficiency in processing live tracking data.

II. How does the process work?

Providers must submit their live tracking data at the same sampling frequency as their system, and to timestamp each message with the collection time and the publication time. Upon receiving the message, test institute (Consumer) will timestamp the message and push it to permanent storage within Google Cloud Platform (GCP).

Various latency values will be calculated as detailed below:

- The "Publication Latency" of the data is assessed by subtracting the collection timestamp from the publication timestamp, both generated by Provider.
- The "Reception Latency" of the data is assessed by subtracting the collection timestamp (generated by Provider) from the reception timestamp (generated by Consumer).
- The "gRPC Latency" is assessed by subtracting the publication timestamp (generated by Provider) from the reception timestamp (generated by Consumer).

Providers are strongly encouraged to adopt the Common Data Format (CDF) — a Standardized Format for Match-Data in Football (Soccer). Details of which can be found later in this document.

This Provider data collected via gRPC will be distributed to Track, the accredited Test Institute for the EPTS Performance Test, in order to analyze the velocity and positional accuracy as per EPTS Test Manual.

III. Connection Details for gRPC Submission

Providers must contact FIFA and the test institute at least 14 days before the start of the EPTS Performance test. Providers will then be given:

- gRPC Server Address
- Authentication Credentials
- Expected Frequency Rate (Hz)

The file format should be according to the requirements in the 'EPTS Data Format Requirements' section of this document.

IV. Testing Prior to Event

Testing gRPC on-site is highly encouraged to minimize troubleshooting once the collection event starts.

To that end, once connection details have been provided – we encourage Providers to test their gRPC publishing whenever convenient.

Test institute will then validate latency and coordinate any issues with the Provider.

V. EPTS Data Format Requirements

Each message should be provided as a JSON string format, such that it can be decoded using *json.loads()* in a Python 3 environment.

More detailed specifications follow:

V.I. Data required in each message

Time of collection

- Name: collected_at
- Description: Timestamp (in UTC) of the data collection
- Format: Float

Time of publication

- Name: published_at
- Description: Timestamp (in UTC) of the data publication
- Format: Float

Frame

- Name: frame_id
- Description: Unique frame identifier
- Format: Integer

Frequency rate

- Name: frequency_rate
- Description: Expected frequency rate in Hz, so we can then compare with the cadence observed in *collected_at*
- Format integer

Pitch dimensions

- Name: pitch_dimensions
- Description: Total length of x & y dimensions in system
- Format: [float, float]

V.II. Player Data

We strongly encourage Providers to adhere to the formats outlined in the **Common Data Format (CDF)**.

The detailed data formats can be found here:

Root	Field	Description	Type
	frame_id	Unique frame identifier	Integer
	period	Period of the match (first_half, second_half, first_half_extra, second_half_extra, shootout)	String
match	id	Unique match identifier	String
teams/home	id	Unique identifier for the home team*	String
teams/away	id	Unique identifier for the away team*	String
teams/players	id	Unique identifier for a player*	String
teams/players	team_id	Unique identifier for the player's team	String
teams/players	x	x location of the player on the pitch (m)*.	Float
teams/players	y	y location of the player on the pitch (m)*.	Float
ball	x	x location of the ball on the pitch (m).	Float
ball	y	y location of the ball on the pitch (m).	Float
ball	z	z location of the ball on the pitch (m).	Float

Because we require additional values such as collected_at, published_at frequency_rate and pitch_dimensions, there are two possible options while adhering to the **Common Data Format**.

**Please note that if adhering to the Common Data Format is not possible for your organization in this timeframe, reach out to the test institute and FIFA to discuss the specifics of your data format.*

Option 1: EPTS required fields included at CDF root level:

```
{
        “published_at”: 123456789.10,
        “collected_at”: 123456789.10,
        " frame_id ": 0 ,
        “frequency_rate”: 50,
        “pitch_dimensions”: (100, 100)
        " period ": " f i r s t _ h a l f " ,
        " match ": {" id ": "74 e6661c "} ,
        " teams ": {
         " home ": {
                  " id ": "1c4abab2025 " ,
                  " players ": [{" id ": "61a2b9ba " , " x ": 0.13 " y ": 4.04} ,
                           ...
                  ]
                  } ,
                  " away ": {
                           " id ": "3f029694 " ,
                           " players ": [{" id ": "abc1234 " , " x ": 0.76 " y ": 13.37} ,
                                    ...
                           ]
                  }
        } ,
        " ball ": {
         " x ": 0.01 ,
         " y ": 23.10 ,
         " z ": 0.33 ,
         ...
        }
}
```

Option 2: CDF root level data nested in “data” parameter:

```
    {
            “published_at”: 123456789.10,
            “collected_at”: 123456789.10,
            " frame_id ": 0 ,
            “frequency_rate”: 50,
            “pitch_dimensions”: (100, 100)
            “data”: {
                “frame_id”: 0,
                " period ": " f i r s t _ h a l f " ,
                " match ": {" id ": "74 e6661c "} ,
                " teams ": {
                           " home ": {
                                    " id ": "11c4abab2025 " ,
                                    " players ": [{" id ": "61a2b9ba " , " x ": 0.13 " y ": 4.04} ,
                                             ...
                                    ]
                           } ,
                           " away ": {
                                    " id ": "3 f029694 " ,
                                    " players ": [{" id ": " abc1234 " , " x ": 0.76 " y ": 13.37} ,
                                             ...
                                    ]
                           }
                } ,
                " ball ": {
                           " x ": 0.01 ,
                           " y ": 23.10 ,
                           " z ": 0.33 ,
                           ...
                }
            }
    }
```